

Real Estate Investment Conditions for Local Citizens Across Europe between 2016 and 2025

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1 Motivation

Real estate is the primary wealth-building tool for most European households, yet property prices are increasingly detaching from local wages and general inflation. Pinpointing exactly how and where this affordability gap widens is vital for understanding regional wealth inequality and identifying where it remains economically viable for local citizens to invest in their own housing markets.

2 Data Description

Three tables were downloaded from Eurostat database, one was HICP - Inflation rate which had historical data of past 10 years and contained 37 different countries and Euro areas. The second was HPI – House price index with historical data of past 10 years and 38 different countries and Euro areas and the third was LCI – Labour cost index containing data from past 10 years and 39 different countries and Euro areas.

A detailed summary of the three raw datasets retrieved from the Eurostat database is provided below, including their source locations, structural dimensions, and key variable definitions prior to the data cleaning process:

- **Dataset 1: Labour Cost Index by NACE Rev. 2 Activity (Nominal Value, Annual Data)**

- **Source:** https://ec.europa.eu/eurostat/databrowser/view/lc_lci_r2_a/default/table?lang=en&category=labour.lc.lci
- **Dimensions:** 5,380 rows $\times$ 11 columns
- **Key Variables & Units:**
 - geo:** Country (Categorical identifier).
 - TIME.PERIOD:** Reference Year, spanning 2016–2025 (Temporal).
 - nace_r2:** Economic sector. Filtered to “B-S” to represent the total business economy.
 - lcstruct:** Cost component. Filtered to “Wages and salaries (total)” to isolate employee income from employer tax burdens.
 - OBS.VALUE:** Hourly labour cost. Measured as the percentage change on the previous period (%).

- **Dataset 2: House Price Index (2015 = 100) - Annual Data**

- **Source:** https://ec.europa.eu/eurostat/databrowser/view/prc_hpi_a/default/table?lang=en&category=prc.prc_hpi.prc_hpi_inx
- **Dimensions:** 3,180 rows $\times$ 10 columns

– **Key Variables & Units:**

geo: Country (Categorical identifier).

TIME_PERIOD: Reference Year, spanning 2016–2025 (Temporal).

purchase: Property type. Filtered to “Total” to include both new and existing residential properties.

OBS_VALUE: Residential property price. Measured as the annual average rate of change (%).

• **Dataset 3: Harmonised Index of Consumer Prices (HICP) - Inflation Rate**

– **Source:** <https://ec.europa.eu/eurostat/databrowser/view/tec00118/default/table?lang=en>

– **Dimensions:** 357 rows $\times$ 10 columns

– **Key Variables & Units:**

geo: Country (Categorical identifier).

TIME_PERIOD: Reference Year, spanning 2016–2025 (Temporal).

coicop18: Consumption category. Filtered to “Total” to capture general consumer inflation.

OBS_VALUE: Consumer price inflation. Measured as the annual average rate of change (%).

3 Cleaning Narrative

The raw datasets downloaded from Eurostat contained large amounts of unnecessary information and numbers of unwanted metadata which had to be cleaned and prepared for proper analysis

Initially, unwanted metadata columns (such as DATAFLOW, OBS_FLAG) were removed to prevent conflicts when merging tables. Wanted variables were renamed from Eurostat’s original names (e.g. geo, TIME_PERIOD) to more easily understandable formats (country, year) to ensure clarity and simple usability.

Provided data were divided into multiple sub-sectors, which would conflict the final averages if not addressed. Multiple filters were applied to each dataset to create a single and comparable variable for each country and year.

House Price Index (HPI) was filtered to “Total” purchases instead of separating new and existing properties as provided in original datasets, utilizing the “Annual average rate of change” to reflect percentage changes each year. Labour Costs were narrowed to the classification “B-S” which stands for the whole economy and only “wages and salaries” were selected to accurately capture workers income and were measured in “Percentage change on previous period”. Inflation (HICP) was standardized to the “Annual average rate of change” to match the percentage growth format as the previous datasets.

Before merging, the intersection of countries across all three datasets was found and others were dropped. The total amount of common countries was 29. The datasets were then combined into one table for easier analysis and a new indicator “Affordability Gap” was added, calculated by subtracting wage growth from HPI.

Lastly, countries missing any information were detected and removed so the results were not inaccurate because of insufficient data for certain countries. From HPI dataset a total of 13 countries was removed, from HICP it was 12 and dataset containing labour costs was reduced by 14 countries, making the final number of studied countries 25.

4 Descriptive Statistics

For a better understanding of the data, a descriptive summary of all key indicators was created (HPI, Wage Growth, Inflation Rate, Affordability Gap). When comparing the central tendencies, a notable skewness was observed. For wage growth, the mean (5.65%) is visibly higher than the median (4.25%), indicating a right-skewed distribution. This means the overall average wage growth is pulled up by a few high-performing years in specific countries. The inflation rate mean (3.31%) and median (2.30%) follow the same trend. The affordability gap shows a lower mean (0.89%) compared to the median (1.10%), which is driven by occasional years of extraordinary wage growth that significantly outpaced the housing market. The HPI mean (6.55%) and median (6.45%) differences show that the HPI is mostly symmetrical.

To better visualize the data scattering as well as maximums and minimums, boxplots for all the key indexes were created. The boxplots show that the inflation rate and wage growth IQRs are relatively narrow, showing the growth being relatively stable. However, both display significant upwards outliers, reaching 19.4% for inflation and 39.6% for wage growth. In contrast, HPI and the affordability gap show greater variance, both having a wider IQR and outliers on both ends of the spectrum. The affordability gap is particularly volatile, fluctuating from an extreme minimum of -32.8% to a maximum of 16.2%.

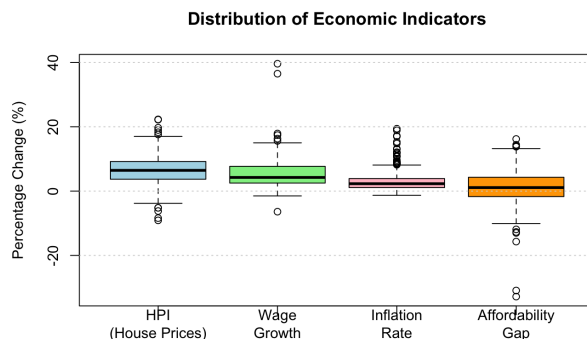


Figure 1: Boxplots for each indicator

To better understand how often these extreme events occur, the data was divided into four percentage brackets: Negative ($\leq 0\%$), Moderate (0-5%), High (5-10%), and Extreme ($\geq 10\%$). As it turned out, the inflation rate remained stable across most countries and time periods, falling into the moderate zone 74.0% of the time, with extreme inflation shocks occurring in just 6.0% of cases. It was also observed that HPI reached the extreme growth zone in 19.6% of cases, compared to 15.2% for wage growth. This indicates that real estate prices are more prone to sharp and rapid surges than the labor market. Finally, when examining the affordability gap, it is evident that in 40.4% of instances, housing became more affordable relative to wages. However, in the remaining 59.6% of cases, the situation was the opposite. Notably, 4.4% of the observations represented an extreme situation, meaning housing prices grew 10 percentage points or more than local wages in a single year.

Subsequent analysis examined the means and volatility (standard deviation) across the 25 observed countries. The data reveals significant differences between observed regions. Countries such as Portugal, the Netherlands, and Spain suffer from the most severe affordability gaps, with house prices outpacing wages by an average of 6.02, 4.63, and 3.72 percentage points per year. On the other hand, countries like Romania (-9.40) and Lithuania (-3.16) exhibited a negative gap, meaning that local wage growth outpaced property prices. However, a closer examination of the standard deviation reveals the instability of those markets. For instance, while Romania shows a highly favorable average gap, it also experiences extreme volatility ($SD = 8.55$). Considering this, there are other markets that offer a negative gap without such high volatility. For example, Bulgaria ($AG = -1.58$; $SD = 2.66$) and Italy ($AG = -0.21$; $SD = 1.81$) may not have the most drastic negative gaps, but they offer some of the lowest standard deviations, making them much more stable.

Finally, on a global scale, the data confirms that real estate prices ($SD = 5.07$) are more volatile than the labor market ($SD = 4.93$), making the resulting Affordability Gap the most unstable metric overall ($SD = 5.96$).

5 Visualization

To provide a clear visualization of the data and make concrete findings, a histogram showing the distribution of the affordability gap, a scatterplot comparing wage growth and HPI, and a Bar chart of average affordability gap by country were created.

An examination of the histogram reveals a clear left skewness of affordability gap with the bulk of the observations falling in the positive, including the highest peak. This demonstrates that in most cases, the house price inflation outpaced local wage growth, which confirms that for most observed countries a decrease in real estate affordability is the most typical outcome.

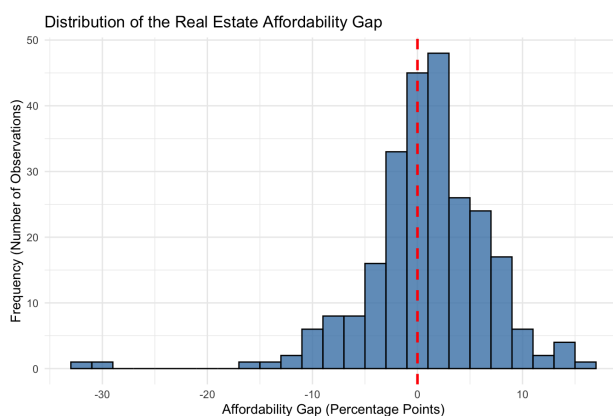


Figure 2: Distribution of Affordability gap

The scatterplot illustrates the relationship between annual wage growth and changes in house prices. The trendline displays a weak positive correlation indicating that when wages rise, house prices also rise. However, within the most common annual wage growth bracket (between 0 and 10 percent), there is a wild fluctuation of house prices. If housing affordability was heavily correlated to wages, the data points would cluster much closer to the trendline. This signifies that the volatility and pricing of housing is largely independent from wages. Additionally, in the descriptive statistics section, it was observed that the mean and median for HPI is relatively similar, which means that while the HPI is volatile, the extremes average out to a symmetrical distribution.



Figure 3: Relationship between Labor income and Real Estate Prices

Lastly, a bar chart was created showing the average affordability gap for each country, providing a clear picture of where exactly it is easiest or hardest to invest in the housing market. Looking at the chart, most countries have a positive average affordability gap, which indicates that in most observed countries, house prices rise faster than wages, making housing less affordable. The situation is worst in Portugal, the Netherlands, Spain and Ireland, where the affordability gap is the highest, meaning in these countries people are losing their purchasing power the fastest.

On the other hand, the green bars at the bottom show the few countries where housing is becoming more affordable. Excluding Italy, Latvia and Poland where the affordability gap is negligible, Bulgaria, Finland, Lithuania and Romania remain as the sole countries where housing is becoming more affordable, with Romania being the biggest negative outlier. However, as previously mentioned, Romania has a large deviation of affordability gap. While on average becoming more affordable, the market suffers from high volatility.

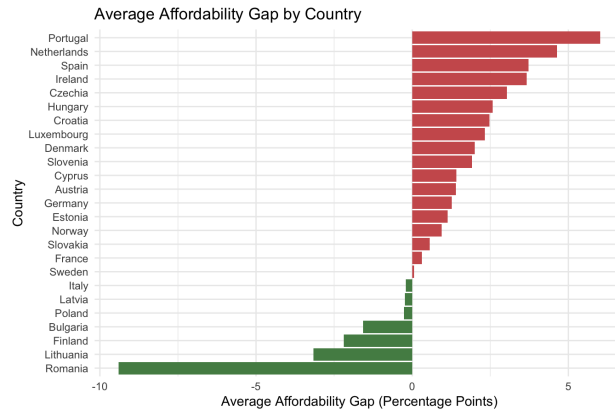


Figure 4: Affordability Gap per Country

6 Central Findings

To answer the main question regarding the detachment of real estate from local economies, multiple correlation tests were conducted. The initial expectation was that property prices are increasingly disconnecting from local wages and general inflation and the statistical evidence supports this hypothesis.

First, a correlation test between wage growth and the House Price Index (HPI) showed a weak positive correlation of 0.29. This compared with the visual evidence from the scatterplot further indicates that the housing market is indeed largely detached from the labor market.

Consequently, to ensure that this divergence is not merely a result of general cost-of-living increases, the inflation rate was tested against HPI, resulting in another weak correlation of 0.32. This demonstrates that real estate prices are not simply tracking normal inflation, but behave rather independently as highly volatile assets.

When combined with earlier descriptive statistics, which showed that housing prices outpaced wages in nearly 60% of observed instances, this highlights that relying solely on labor income is an increasingly unreliable method for European citizens to build wealth through their local housing markets, as real estate costs tend to generally outpace them.

7 Limitations

While the data clearly shows a natural divergence between wages and house prices on the national level, it does not analyze differences on the regional level. There might be differences between rural areas and large cities. Additionally, the LCI only tracks labor costs meaning the data completely omits the purchasing power of entrepreneurs and self-employed individuals. Furthermore, the analysis only proves correlation, not causation. In order to prove causation, a larger time period would have to be studied including local housing supply, foreign investment as well as mortgage interest rates.

8 Collaboration

In order to collaborate on the code, we used Git. The report was initially drafted in a shared Word document to allow for continuous team input and discussion, before being finalized in LaTeX. Although the entire project, including data interpretation and code reviews, was a joint team effort, the primary responsibilities were divided as follows:

- **Nikola Jankovič**
 - GitHub setup
 - Visualization
 - LaTeX copy and stylistic editing
 - Limitations
 - Central findings
- **Josef Mitošinka**
 - Finding datasets
 - Cleaning Narrative
 - Data Description
 - Slides
- **Ondra Čermák**
 - Motivation
 - Data Description
 - Descriptive Statistics
 - Limitations